

Applications of Definite Integrals

Area between Two curves:

Let $y_1 = f_1(x)$ and $y_2 = f_2(x)$ are continuous for $a \leq x \leq b$, and $f_1(x) \geq f_2(x)$ for $a \leq x \leq b$

$\therefore$ The area between two curves is

$$\int_a^b [f_1(x) - f_2(x)] dx.$$

Ex. 1 Find the area bounded by the parabola $y = 2 - x^2$ and the straight line $y = -x$

Sol:

we solve $2 - x^2 = -x$

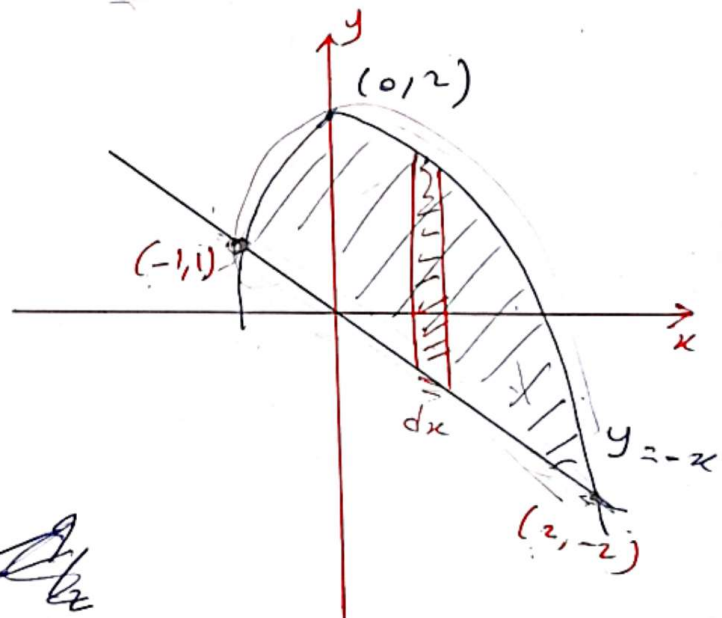
$$\Rightarrow x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$x = 2, x = -1$$

$$A = \int_{-1}^2 (2 - x^2 + x) dx$$

$$= \left[2x - \frac{1}{3}x^3 + \frac{1}{2}x^2 \right]_{-1}^2 = \frac{9}{2}$$



$$[\quad] - [\quad]$$

Ex. 2: Find the area bounded on the right by the line $y = x - 2$ on the left by the parabola $x = y^2$ and below by the x -axis

Sol: we solve

$$y^2 = y + 2 \rightarrow y^2 - y - 2 = 0$$

$$(y - 2)(y + 1) = 0$$

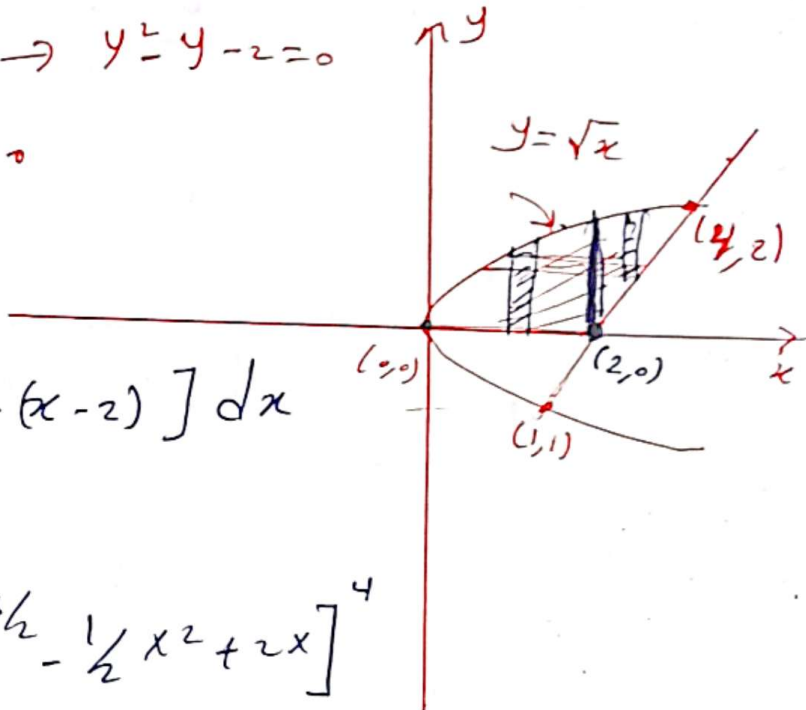
$$\therefore y = 2 \text{ or } y = -1$$

$$A = \int_0^2 \sqrt{x} dx + \int_2^4 [\sqrt{x} - (x - 2)] dx$$

$$= \left. \frac{2}{3} x^{3/2} \right|_0^2 + \left[\frac{2}{3} x^{3/2} - \frac{1}{2} x^2 + 2x \right]_2^4$$

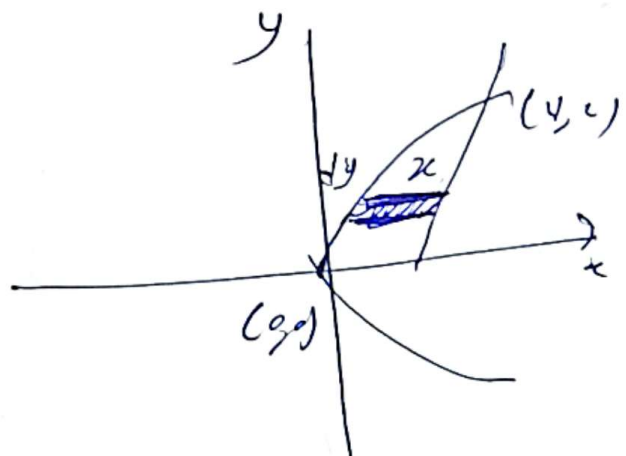
$$= \frac{2}{3} (2)^{3/2} + \left[\left(\frac{2}{3} (4)^{3/2} - \frac{1}{2} (4)^2 + 2(4) \right) - \left(\frac{2}{3} (2)^{3/2} - \frac{1}{2} (2)^2 + 2(2) \right) \right]$$

$$= \frac{10}{3}$$



أو يمكننا حلها بالنسبة لـ y

$$A = \int_0^2 [y + 2 - y^2] dy$$



(3)

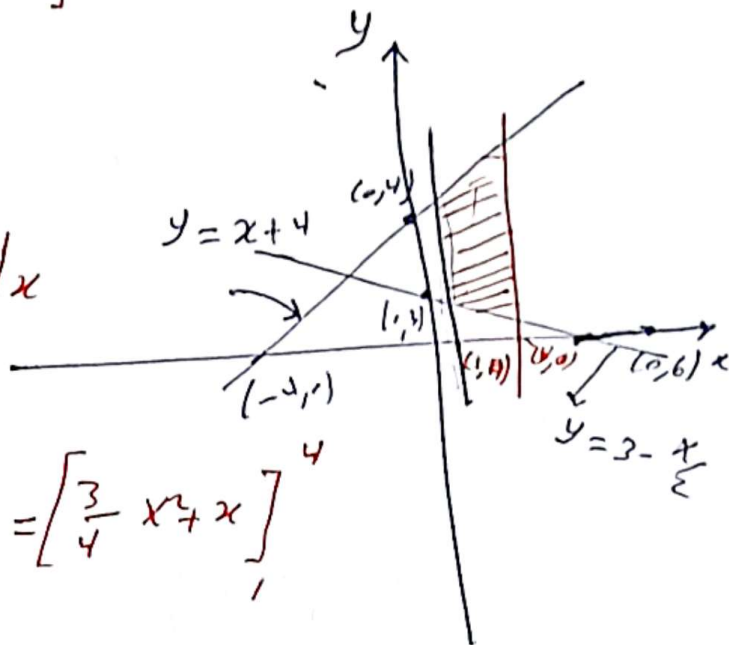
Ex. 3: Find the area bounded by the curve $y = x + 4$ and the curve $y = 3 - \frac{x}{2}$ over the interval $[1, 4]$

Sol:

$$A = \int_1^4 \left[x + 4 - 3 + \frac{x}{2} \right] dx$$

$$= \int_1^4 \left[\frac{3}{2}x + 1 \right] dx = \left[\frac{3}{4}x^2 + x \right]_1^4$$

$$= \frac{57}{4}$$



Ex. 4: Find the area bounded by the curve $y = 9 - \frac{x^2}{4}$ and above the curve $y = 6 - x$

Sol:

~~The~~ we write the Eq. $y = 9 - \frac{x^2}{4}$ in the form

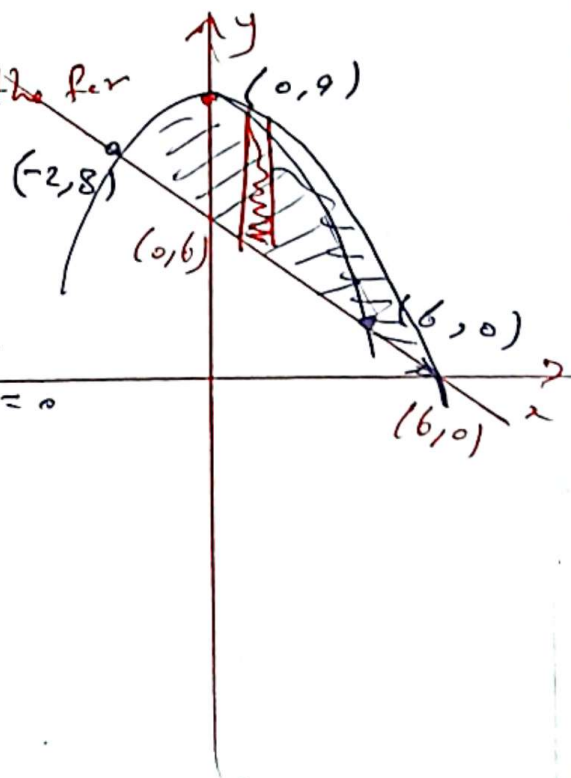
$$x^2 = -4(y - 9)$$

We solve $6 - x = 9 - \frac{x^2}{4}$

$$x^2 - 4x - 12 = 0 \rightarrow (x - 6)(x + 2) = 0$$

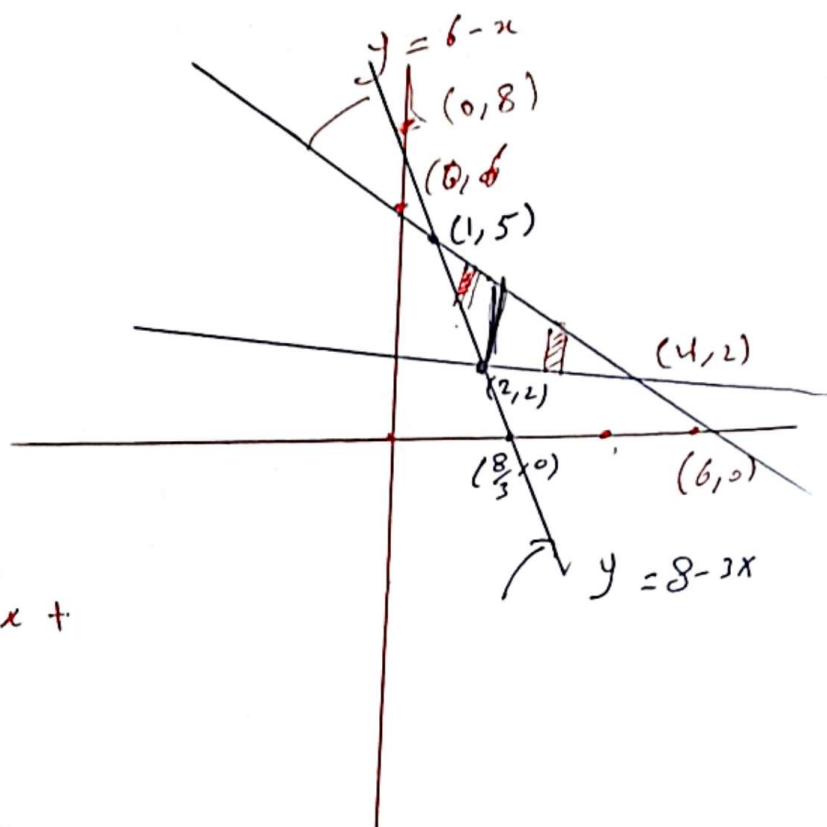
$$A = \int_{-2}^6 \left[\left(9 - \frac{x^2}{4} \right) - (6 - x) \right] dx$$

$$= \frac{64}{3}$$



Q. Find the area bounded by $y = 8 - 3x$, $y = 6 - x$
and $y = 2$

Sol.



$$A = \int_0^2 [6 - x - 8 + 3x] dx +$$

$$\int_2^4 [6 - x - 2] dx$$

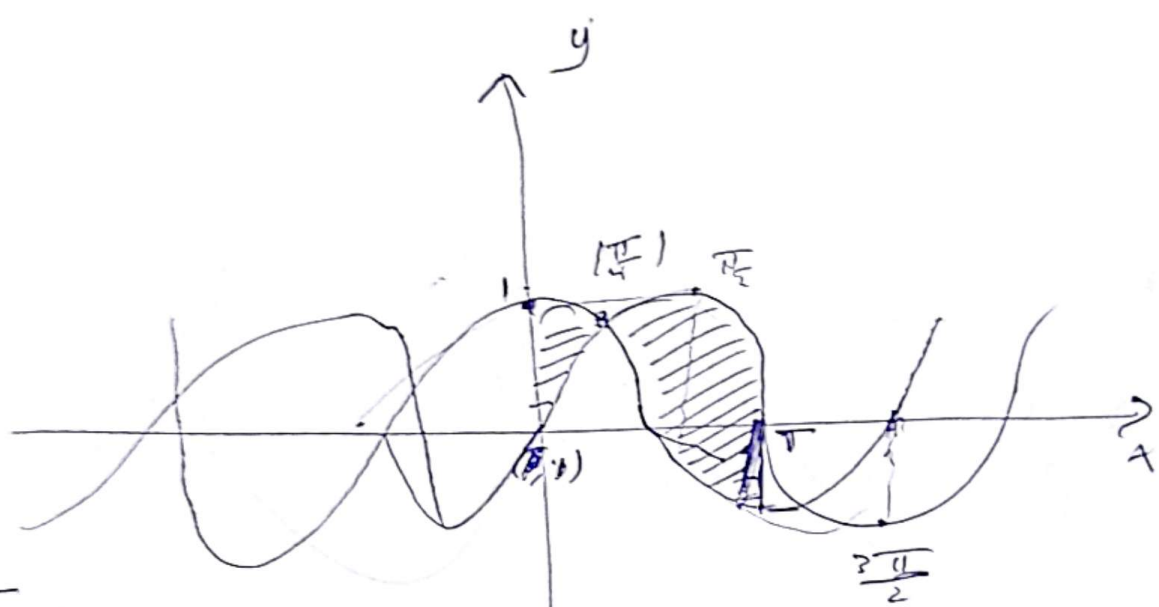
$$= 3$$

=

(2)

Ex 3: If R is the region between the graphs of functions $f(x) = \sin x$ and $g(x) = \cos x$ over the interval $[0, \pi]$ find the area of R

Sol.:



$$\begin{aligned} A &= \int_0^{\pi} [f(x) - g(x)] dx \\ &= \int_0^{\frac{\pi}{4}} [\cos x - \sin x] dx + \int_{\frac{\pi}{4}}^{\pi} [\sin x - \cos x] dx \\ &= 2\sqrt{2} \end{aligned}$$

Ex. ② Find the area of the region R bounded by the curves $y = x^3$ and $y = x^2$

Sol: Find the intersect points

let

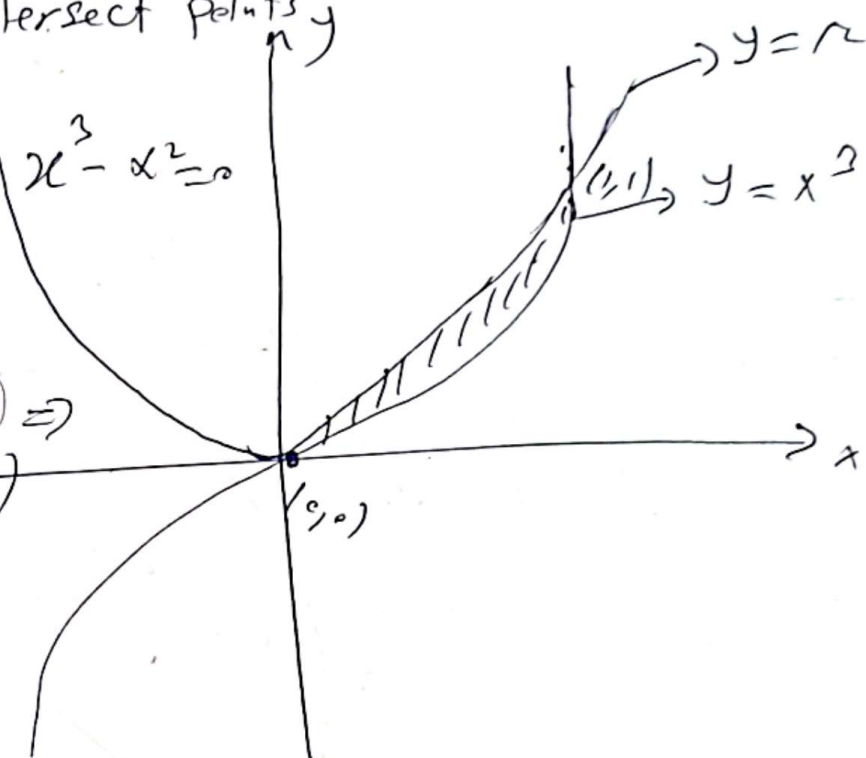
$$x^3 = x^2 \Rightarrow x^3 - x^2 = 0$$

$$\Rightarrow x^2(x-1) = 0$$

$$\Rightarrow x = 0, x = 1$$

$$(0,0), (1,1)$$

$$A = \int_0^1 [x^2 - x^3] dx$$



Ex. 1 Find the area of the region R bounded by the curves $y = x^3$ and the line $y = 9x$, $x \geq 0$

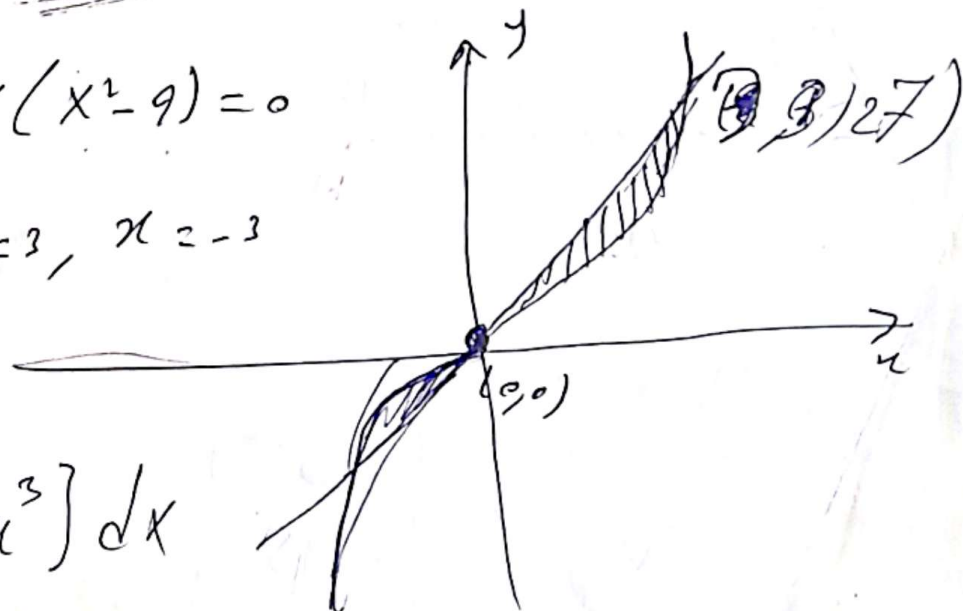
Sol: let

$$x^3 = 9x \rightarrow x(x^2 - 9) = 0$$

$$\Rightarrow x = 0, x = 3, x = -3$$

$$A = \int_{-3}^3 [f(x) - g(x)] dx$$

$$\therefore = \int_0^3 [9x - x^3] dx$$



① Definite integral

If $\int_a^x f(x) dx = g(x) + C$

Then $\boxed{\int_a^b f(x) dx = g(b) - g(a)}$

Properties of definite integrals

① $\int_a^b f(x) dx = \int_a^b f(t) dt = \dots = \int_a^b f(z) dz$

② $\int_a^b f(x) dx = - \int_b^a f(x) dx$ ③ $\int_a^a f(x) dx = 0$

④ $\int_a^b c f(x) dx = c \int_a^b f(x) dx$

⑤ $\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$

⑥ $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, $a < c < b$

⑦ $\int_0^a f(x) dx = \begin{cases} 0 & \text{if } f(x) \text{ is odd} \\ 2 \int_0^a f(x) dx & \text{if } f(x) \text{ is even} \end{cases}$

⑧ $\int_0^a f(a-x) dx = \int_0^a f(x) dx$

(2)
Examples

Find the following integrals!

Ex. ①

$$\textcircled{1} \int_0^5 x^2 dx$$

$$\underline{\text{sol:}} \int_0^5 x^2 dx = \left[\frac{1}{3} x^3 \right]_0^5 = \frac{5^3}{3} - 0 = \frac{125}{3}$$

Ex.

$$\textcircled{2} \int_a^b e^x dx =$$

$$\underline{\text{sol:}} \int_a^b e^x dx = \left. e^x \right|_a^b = e^b - e^a$$

Ex.

$$\textcircled{3} \int_0^a \sqrt{a^2 - x^2} dx$$

$$\underline{\text{sol}} = \text{let } x = a \sin u \rightarrow dx = a \cos u du$$

$$\text{at } x = a \rightarrow a = a \sin u \rightarrow \sin u = 1 \rightarrow u = \frac{\pi}{2}$$

$$\text{at } x = 0 \rightarrow 0 = a \sin u \rightarrow u = 0$$

$$\begin{aligned} \text{then } \int_0^a \sqrt{a^2 - x^2} dx &= \int_0^{\frac{\pi}{2}} \sqrt{a^2 - a^2 \sin^2 u} \cdot a \cos u du \\ &= \int_0^{\frac{\pi}{2}} a \cos u \cdot a \cos u du = a^2 \int_0^{\frac{\pi}{2}} \cos^2 u du \\ &= a^2 \int_0^{\frac{\pi}{2}} \frac{1}{2} [1 + \cos 2u] du \end{aligned}$$

(3)

$$= \frac{a^2}{2} \int_0^{\frac{\pi}{2}} [1 + \cos u] du$$

$$= \frac{a^2}{2} \left[u + \frac{1}{2} \sin 2u \right]_0^{\frac{\pi}{2}} = \frac{a^2}{4} \pi$$

Ex:Ex. (4)

$$\int_{-5}^5 \frac{x}{\sqrt{3+x^2}} dx$$

Sol: $\therefore$ The function $\frac{x}{\sqrt{3+x^2}}$ odd

$$\text{Then } \int_{-5}^5 \frac{x}{\sqrt{3+x^2}} dx = 0$$

Ex:5:

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3 x dx$$

Sol:

$$\begin{aligned} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3 x dx &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 x \cos x dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [1 - \sin^2 x] \cos x dx \\ &= \left[\sin x - \frac{\sin^3 x}{3} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \end{aligned}$$

(4)

Then

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3 x \, dx = \left[1 - \frac{1}{3}\right] - \left[-1 + \frac{1}{3}\right]$$

$$= 2 \left[1 - \frac{1}{3}\right] = \frac{4}{3}$$

Ex 6: $\int_0^{\frac{\pi}{2}} \sin^2 x \, dx =$

Sol: $\int_0^{\frac{\pi}{2}} \sin^2 x \, dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} [1 - \cos 2x] \, dx$

$$= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left\{ \left[\frac{\pi}{2} \right] - 0 \right\} = \frac{\pi}{4}$$

Ex. 7: $\int_1^2 (x^3 + 3x^2 + 1) \, dx$

$$= \left[\frac{1}{4} x^4 + x^3 + x \right]_1^2 = \left[\frac{16}{4} + 8 + 2 \right] - \left[\frac{1}{4} + 1 + 1 \right]$$

$$= \frac{56}{4} - \frac{9}{4} = \frac{47}{4}$$

Ex: 8: $\int_0^{\frac{\pi}{2}} x \sin 2x \, dx$ (5)

Sol:

Let $u = x$

$du = dx$

$dv = \sin 2x \, dx$

$v = -\frac{1}{2} \cos 2x$

$$\therefore \int_0^{\frac{\pi}{2}} x \sin 2x \, dx = \left[-\frac{x}{2} \cos 2x + \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos 2x \, dx \right]$$

$$= \left[-\frac{x}{2} \cos 2x \right]_0^{\frac{\pi}{2}} + \frac{1}{4} \left[\sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= \left[-\frac{\frac{\pi}{2}}{2} (-1) - 0 \right] + \frac{1}{4} [0]$$

$$= \left[\frac{\pi}{2} \right]$$

Ex. 9: $\int_0^3 \frac{2x-1}{(x+1)(x-2)} dx$ (6)

Sol:

$$\therefore \frac{2x-1}{(x+1)(x-2)} = \frac{A}{x+1} + \frac{B}{x-2}$$

$$\Rightarrow -2x-1 = A(x-2) + B(x+1)$$

at $x = -2 \rightarrow B = 1$

at $x = -1 \rightarrow A = 1$

$$\therefore \frac{2x-1}{(x+1)(x-2)} = \frac{1}{x+1} + \frac{1}{x-2}$$

$$(i.e) \int_0^3 \frac{2x-1}{(x+1)(x-2)} dx = \int_0^3 \frac{1}{x+1} dx + \int_0^3 \frac{1}{x-2} dx$$

$$= \ln(x+1) \Big|_0^3 + \ln(x-2) \Big|_0^3$$

$$= [\ln 4 - \ln(1)] + [\ln(1) - \ln(-2)]$$

evaluate

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

$$\therefore \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx$$

$$= \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$$

$$= \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx - \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

$$\Rightarrow \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

$$= -\frac{\pi}{2} \int_0^{\pi} \frac{d(\cos x)}{1 + \cos^2 x}$$

$$= -\frac{\pi}{2} \left\{ \tan^{-1}(\cos x) \right\}_0^{\pi}$$

$$= -\frac{\pi}{2} \left\{ \tan^{-1}(\cos \pi) - \tan^{-1}(\cos 0) \right\}$$

$$x: I = \int_0^{1/2} \frac{\sqrt{1-x}}{\sqrt{1+x}} dx$$

sol:

$$I = \int_0^{1/2} \frac{\sqrt{1-x}}{\sqrt{1+x}} \times \frac{\sqrt{1-x}}{\sqrt{1-x}} dx$$

$$= \int_0^{1/2} \frac{1-x}{\sqrt{1-x^2}} dx$$

$$= \int_0^{1/2} \frac{1}{\sqrt{1-x^2}} dx - \int_0^{1/2} x(1-x^2)^{-1/2} dx$$

$$= \sin^{-1} x \Big|_0^{1/2} + \frac{1}{2} (1-x^2)^{1/2} \Big|_0^{1/2}$$

$$= \left[\sin^{-1} \left(\frac{1}{2} \right) - \sin^{-1}(0) \right] + \left[\left(1 - \frac{1}{4} \right)^{1/2} - 1 \right]$$

